

FELIPE ANTONIO FUMERO MORA

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Portfolio: felipefumero.com | GitHub: github.com/ffumero2003

PROFILE

Data Engineer focused on building data pipelines and AI-powered data products that turn raw data into decisions. Ships end to end, from ingestion and modeling to natural-language query and explanation layers, on a strong Python and SQL foundation, with deep AI/LLM integration experience (18 Anthropic certifications) and a full-stack background that has put real products into production. C1 English.

PROFESSIONAL EXPERIENCE

CriticalRiver — Costa Rica (US AI & data services consulting firm)

Junior Data Engineer | March 2026 – May 2026

- Selected for CriticalRiver Data & AI Freshers Program — structured DE training in Airflow, dbt, BigQuery, Snowflake, Power BI, Tableau, SQL
- Built end-to-end GCP practices (Superstore dataset): GCS setup, BigQuery loading with partitioning/clustering, SQL cost optimization, Looker Studio dashboard; configured an Airflow weather pipeline in WSL 2 (Docker, PostgreSQL)
- Scored 80% on the formal WeCP Data Engineering assessment (Airflow, dbt, SQL, Python, BI tools, Agile)

PayFacility — Costa Rica (fintech: payment processing & POS for LATAM)

Software Developer Intern | October 2025 - January 2026

- Worked with DS Spread application in sandbox environment to learn payment transaction workflows.
- Conducted extensive transaction testing and debugging using ADB/Logcat to understand authorization, reversal, and error handling processes

Social Brands – Costa Rica (authorized retailer, Northweek/Hawkers eyewear)

Front-End Developer | October 2024 - September 2025

- Built e-commerce website from scratch using Shopify Liquid, HTML/CSS, and JavaScript
- Implemented strategic upselling features, promotional displays, and optimized navigation to enhance customer journey and increase sales potential
- Delivered production-ready solution independently, managing all development phases

Link: <https://northweek.cr/>

KEY PROJECTS

GitHub Firehose — Developer Activity Intelligence | Real-Time Dev Activity Pipeline at Scale

PySpark, BigQuery, dbt, Apache Airflow, GitHub Archive, Looker Studio, SQL

- Built an end-to-end PySpark pipeline ingesting 100M+ nested JSON events from the GitHub Archive firehose — a volume and nesting depth beyond what pandas can handle — using a hand-written explicit schema for contract safety and filter-early predicate pushdown
- Transformed and exploded the nested commits array into a separate commit-grain fact (a deliberate grain change), writing a BigQuery star schema (2 facts + 4 dimensions) via the direct Storage Write API
- Modeled 5 analytical marts with dbt and enforced 11 data-quality tests (unique, not_null, referential integrity) reconciling exactly against the Spark layer's counts
- Orchestrated idempotent daily ingestion as an Apache Airflow DAG (download → Spark → dbt run → dbt test), halting on any failure

Code: github.com/ffumero2003/github-firehose-activity

Credit Risk Scoring + Explainable AI – End-to-End Credit Pipeline with FCRA(Fair Credit Reporting

Act)-Style AI Explanations Python, BigQuery, dbt, XGBoost, SHAP, Claude API, Apache Airflow, Docker, React, Flask, Looker Studio

- Built an end-to-end credit-risk scoring pipeline over ~308K loan applications and ~1.7M bureau records, scoring per-applicant default risk with an XGBoost model (vs. logistic-regression baseline; AUC/Gini/KS metrics)
- Used SHAP to generate per-applicant reason codes, then a Claude-powered layer turning them into FCRA-style adverse-action notices, excluding protected attributes for ECOA fair-lending alignment
- Orchestrated the full ingest, transform, score, and quality-check flow as an Airflow DAG in Docker

Code: github.com/ffumero2003/credit-risk-explainable-ai

Connector Service — Caching API Gateway | Resilient Integration "Glue" Service

Python, Django, Django REST Framework, Redis, Docker, REST APIs

- Built a caching API gateway in Django REST Framework that fronts an upstream public API and returns a stable response contract, decoupling consumers from upstream changes
- Implemented cache-aside caching with Redis (TTL expiry) so a cache hit skips the upstream entirely — cutting redundant calls and respecting the upstream's rate limit

- Added a resilience layer: bounded request timeouts, retry with exponential backoff on transient failures, and correct HTTP status codes (404 / 502 / 503) for graceful degradation
- Self-directed weekend build mapping Flask concepts to Django/DRF (routes → URL dispatcher, view functions → APIView, jsonify → Response)

Code: github.com/ffumero2003/django-connector-service

LATAM Economic Intelligence Dashboard – End-to-End Data Pipeline with AI Query Layer Python, pandas, BigQuery, dbt, Looker Studio, Claude API, React, Flask

- Built an end-to-end pipeline ingesting live U.S. Federal Reserve (FRED) indicators into BigQuery (retry/backoff for API limits) and modeled them into analytical tables with dbt
- Developed a natural-language query layer (Claude API) converting plain-English questions into read-only SQL against BigQuery, returning explained results

Code: github.com/ffumero2003/latam-economic-intelligence-dashboard

PuffZero – Full-Stack Mobile App (Live on Android | iOS — In Review)

React Native, Expo, Supabase, TypeScript, Google Auth, AI Integration

- Built a full-stack vaping cessation app (iOS/Android, 108+ source files, MVVM) with complete Supabase auth (email/password, Google Sign-In, verification, password reset) and an OpenAI-powered AI chat assistant
- Implemented notifications, progress tracking with chart visualizations, and paywall integration via RevenueCat/Stripe

Live: <https://puffzero-landing.vercel.app/>

And more projects in my portfolio website: felipecumero.com

TECHNICAL SKILLS

Data Engineering: SQL, Python, PySpark, pandas, BigQuery, Power BI, Tableau, Airflow, dbt, Snowflake

Backend & Databases: Python, Flask, Django, Django REST Framework, PostgreSQL, Redis, Node.js, Express.js, REST APIs, Google Cloud Storage

Frontend: React, React Native, TypeScript, JavaScript, HTML5, CSS3, Tailwind

Tools: Git/GitHub, Docker, WSL 2, Azure DevOps, Postman, Looker Studio, Vite, Vercel, Figma

AI/LLM Integration: Claude API, Claude Code, Model Context Protocol (MCP), AI Agents, Prompt Engineering

AI-Assisted Development: Claude, Cursor, Gemini, ChatGPT

ADDITIONAL TRAINING

AI & Claude Development (Anthropic, 2026):

- 18 official certifications (Skilljar) — Claude API & Claude Code, Model Context Protocol (servers + integrations), AI Agents/Subagents/Skills, Bedrock & Vertex AI deployment, AI Fluency.

Credentials: [linkedin.com/in/felipe-fumero-b5186030b](https://www.linkedin.com/in/felipe-fumero-b5186030b)

Data & Engineering Foundations:

- Python: Introduction + Intermediate (DataCamp, 2023)
- Complete Full-Stack Web Development Bootcamp – front-end focused (2024)

Mentorship:

David Luna (Senior Dev, Novacomp / Professor, Cenfotec) — React architecture, design patterns, advanced hooks, state management (2025).

EDUCATION

Cenfotec University - San Jose, Costa Rica

Software Development Technician | October 2022 - July 2024

Technical certification in full-stack development. Coursework: React, TypeScript, JavaScript, Node.js, databases, software architecture, and modern web development practices.

LEAD University - San Jose, Costa Rica

Bachelor of Business Administration | October 2022 - August 2026

Relevant Coursework: Digital Marketing & E-commerce, Business Strategy, Operations & Services, Investment Analysis

LANGUAGES

- **Spanish** (Native)
- **English** (C1 - Advanced - Professional working proficiency)